Mathematics: analysis and approaches
Standard level
Paper 2
4.1–4.6 Statistics and probability

13 October 2026

W4ss Academy practice set 1 | **Topic** 4.1–4.6 Statistics and probability | **Ref.** W4SS–MAA–T4–P2–01

Student name

30 minutes

Instructions to students

- Write your name in the boxes above.
- Do not open this examination paper until instructed to do so.
- A graphic display calculator is required for this paper.
- Section A: answer all questions. Answers must be written within the answer boxes provided.
- Section B: answer all questions on the answer pages at the end of this paper.
- Unless otherwise stated in the question, all numerical answers should be given exactly or correct to three significant figures.
- A clean copy of the **mathematics: analysis and approaches formula booklet** is required for this paper.
- The maximum mark for this examination paper is **[27 marks]**.

Full marks are not necessarily awarded for a correct answer with no working. Answers must be supported by working and/or explanations. Solutions found from a graphic display calculator should be supported by suitable working. For example, if graphs are used to find a solution, you should sketch these as part of your answer. Where an answer is incorrect, some marks may be given for a correct method, provided this is shown by written working. You are therefore advised to show all working.

Section A

Answer **all** questions. Answers must be written within the answer boxes provided. Working may be continued below the lines, if necessary.

1. [Maximum mark: 4]

A data set consisting of 16 test scores has mean 14.5. One test score of 9 requires a second marking and is removed from the data set.

Find the mean of the remaining 15 test scores.

2. [Maximum mark: 4]

The number of hours spent exercising each week by a group of students is shown in the following table.

Exercising time (in hours)	Number of students
2	5
3	1
4	4
5	3
6	x

The median is 4.5 hours.

(a) Find the value of x. [2]

(b) Find the standard deviation. [2]

3. [Maximum mark: 8]

At a school, 70% of the students play a sport and 20% of the students are involved in theatre. 18% of the students do neither activity.

A student is selected at random.

(a) Find the probability that the student plays a sport and is involved in theatre. [2]

(b) Find the probability that the student is involved in theatre, but does not play a sport. [2]

(This question continues on the following page)

(Question 3 continued)

At the school 48% of the students are girls, and 25% of the girls are involved in theatre.

A student is selected at random. Let G be the event "the student is a girl" and let T be the event "the student is involved in theatre".

(c) Find $P(G \cap T)$. [2]

(d) Determine if the events G and T are independent. Justify your answer. [2]

Section B

Answer **all** questions on the answer pages at the end of this paper. Please start each question on a new page.

4. [Maximum mark: 11]

The principal of a high school is concerned about the effect social media use might be having on the self-esteem of her students. She decides to survey a random sample of 9 students to gather some data. She wants the number of students in each grade in the sample to be, as far as possible, in the same proportion as the number of students in each grade in the school.

(a) State the name for this type of sampling technique. [1]

The number of students in each grade in the school is shown in table.

Grade	Number of Students
9	60
10	83
11	33
12	84

(b) (i) Show that 3 students will be selected from grade 12. [3]

(ii) Calculate the number of students in each grade in the sample. [2]

In order to select the 3 students from grade 12, the principal lists their names in alphabetical order and selects the 28th, 56th and 84th student on the list.

(c) State the name for this type of sampling technique. [1]

Once the principal has obtained the names of the 9 students in the random sample, she surveys each student to find out how long they used social media the previous day and measures their self-esteem using the Rosenberg scale. The Rosenberg scale is a number between 10 and 40, where a high number represents high self-esteem.

Student	A	B	C	D	E	F	G	H	I
Time spent on social media, t (hours)	3	1.2	2.5	4.1	4.7	3.6	2.9	1.7	0.5
Self-Esteem, s (Rosenberg Scale)	25	33	26	20	21	22	23	25	31

(d) (i) Calculate Pearson's product moment correlation coefficient, r. [2]

(ii) Interpret the meaning of the value of r in the context of the principal's concerns. [1]

(iii) Explain why the value of r makes it appropriate to find the equation of a regression line. [1]

Do not write solutions on this page.

Section B — answers to question 4

Section B — answers to question 4 (continued)

Sources of the questions

This practice paper was put together by W4ss Academy for internal class use. The questions are taken from the IB DP Mathematics: analysis and approaches Questionbank (standard level, topic 4 Statistics and probability) and cover subtopics 4.1 to 4.6. They remain the property of the International Baccalaureate Organization and must not be published or sold.

Section A — question 1	EXN.2.SL.TZ0.1 · 4.3 mean
Section A — question 2	22M.2.SL.TZ1.2 · 4.3 median and standard deviation of a frequency table
Section A — question 3	21M.2.SL.TZ2.4 · 4.6 combined events, conditional probability, independence
Section B — question 4	EXM.2.SL.TZ0.1, parts (a) to (d)(iii) · 4.1 stratified and systematic sampling · 4.4 correlation coefficient r — part (e), regression line of t on s (4.10), left out